

DBMS

Database Design and Normalization for an Organization Management System

```
CREATE DATABASE OrganDB;
```

```
show databases;
```

```
USE OrganDB;
```

```
CREATE TABLE Employees (  
EmployeeID INT PRIMARY KEY,  
Name VARCHAR(100) NOT NULL,  
DepartmentID INT NOT NULL  
);
```

```
CREATE TABLE Departments (  
DepartmentID INT PRIMARY KEY,  
DepartmentName VARCHAR(100) NOT NULL,  
ManagerID INT  
);
```

```
CREATE TABLE Projects (  
ProjectID INT PRIMARY KEY,  
ProjectName VARCHAR(100) NOT NULL,  
DepartmentID INT NOT NULL  
);
```

```
show tables;
```

```
select * from Employees;
```

```
ALTER TABLE Employees
```

```
ADD FOREIGN KEY (DepartmentID) REFERENCES Departments(DepartmentID);
```

```
ALTER TABLE Projects
```

```
ADD FOREIGN KEY (DepartmentID) REFERENCES Departments(DepartmentID);
```

```
select * from Employees;
```

Output:

The screenshot displays the MySQL Workbench interface. The main editor shows a SQL script with the following content:

```
14 CREATE TABLE Projects (  
15 ProjectID INT PRIMARY KEY,  
16 ProjectName VARCHAR(100) NOT NULL,  
17 DepartmentID INT NOT NULL  
18 );  
19 show tables;  
20 select * from Departments;  
21 ALTER TABLE Employees  
22 ADD FOREIGN KEY (DepartmentID) REFERENCES Departments(DepartmentID);
```

Below the editor, the 'Result Grid' shows data from the 'Departments' table:

DepartmentID	DepartmentName	ManagerID
6123010	bioinformatics	135446
6123011	Bac.Bioinformatics	125789

The 'Output' pane at the bottom shows a log of actions and errors:

#	Time	Action	Message	Duration / Fetch
46	14:28:08	select * from Employees LIMIT 0, 1000	Error Code: 1146. Table 'organdb.employees' doesn't exist	0.000 sec
47	14:28:15	select * from Employees LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
48	14:28:51	select * from Employees LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
49	14:31:21	select * from Department LIMIT 0, 1000	Error Code: 1146. Table 'organdb.department' doesn't exist	0.000 sec
50	14:31:41	INSERT INTO 'organdb'.Employees ('EmployeeID', 'Name', 'DepartmentID') VALUES (1203, 'hema', bioinform...	1366. Incorrect integer value: 'bioinformatics' for column 'DepartmentID' at row 1	0.000 sec
51	14:32:03	select * from Departments LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

The interface also includes a 'Navigator' on the left showing a list of schemas (hemanth, organidb, organizationdb, retail, sakila, school_db, sys, world) and a 'Context Help' pane on the right.